Series ϵFGHE

Set-5

प्रश्न-पत्र कोड
Q.P. Code **65(B)**

रोल नं.
Roll No.

परीक्षार्थी प्रश्न-पत्र कोड को उत्तर-पुस्तिका के मुख-पृष्ठ पर अवश्य लिखें ।
Candidates must write the Q.P. Code on the title page of the answer-book.

गणित

(केवल दृष्टिबाधित परीक्षार्थियों के लिए)

MATHEMATICS

(FOR VISUALLY IMPAIRED CANDIDATES ONLY)

*निर्धारित समय : **3** घण्टे* | *अधिकतम अंक : **80***

*Time allowed : **3** hours* | *Maximum Marks : **80***

- कृपया जाँच कर लें कि इस प्रश्न-पत्र में मुद्रित पृष्ठ **23** हैं ।
- प्रश्न-पत्र में दाहिने हाथ की ओर दिए गए प्रश्न-पत्र कोड को परीक्षार्थी उत्तर-पुस्तिका के मुख-पृष्ठ पर लिखें ।
- कृपया जाँच कर लें कि इस प्रश्न-पत्र में **38** प्रश्न हैं ।
- **कृपया प्रश्न का उत्तर लिखना शुरू करने से पहले, उत्तर-पुस्तिका में प्रश्न का क्रमांक अवश्य लिखें ।**
- इस प्रश्न-पत्र को पढ़ने के लिए 15 मिनट का समय दिया गया है । प्रश्न-पत्र का वितरण पूर्वाह्न में 10.15 बजे किया जाएगा । 10.15 बजे से 10.30 बजे तक छात्र केवल प्रश्न-पत्र को पढ़ेंगे और इस अवधि के दौरान वे उत्तर-पुस्तिका पर कोई उत्तर नहीं लिखेंगे ।
- Please check that this question paper contains **23** printed pages.
- Q.P. Code given on the right hand side of the question paper should be written on the title page of the answer-book by the candidate.
- Please check that this question paper contains **38** questions.
- **Please write down the serial number of the question in the answer-book before attempting it.**
- 15 minute time has been allotted to read this question paper. The question paper will be distributed at 10.15 a.m. From 10.15 a.m. to 10.30 a.m., the students will read the question paper only and will not write any answer on the answer-book during this period.

General Instructions :

Read the following instructions very carefully and strictly follow them :

(i) *This question paper contains **38** questions. **All** questions are **compulsory**.*

(ii) *This question paper is divided into **five** Sections – **A, B, C, D** and **E**.*

(iii) *In **Section A,** Questions no. **1** to **18** are multiple choice questions (MCQs) and questions number **19** and **20** are Assertion-Reason based questions of **1** mark each.*

(iv) *In **Section B,** Questions no. **21** to **25** are very short answer (VSA) type questions, carrying **2** marks each.*

(v) *In **Section C,** Questions no. **26** to **31** are short answer (SA) type questions, carrying **3** marks each.*

(vi) *In **Section D,** Questions no. **32** to **35** are long answer (LA) type questions carrying **5** marks each.*

(vii) *In **Section E,** Questions no. **36** to **38** are case study based questions carrying **4** marks each.*

(viii) *There is no overall choice. However, an internal choice has been provided in 2 questions in Section B, 3 questions in Section C, 2 questions in Section D and 2 questions in Section E.*

(ix) *Use of calculators is **not** allowed.*

SECTION A

This section comprises multiple choice questions (MCQs) of 1 mark each.

1. The domain of the function $\cos^{-1} x$ is :

(a) $[0, \pi]$ (b) $(-1, 1)$

(c) $[-1, 1]$ (d) $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

2. In a 3×3 matrix $A = [a_{ij}]$ whose elements are given by $a_{ij} = \frac{1}{2}|-3i + j|$, the element a_{31} is :

(a) -4 (b) 5

(c) 4 (d) 8

3. If $2\begin{bmatrix} 1 & 3 \\ 0 & x \end{bmatrix} + \begin{bmatrix} y & 0 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 5 & 6 \\ 1 & 8 \end{bmatrix}$, then $2x - y$ is equal to :

(a) 3 (b) 13

(c) -3 (d) 0

4. If $A = \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}$, then A^{-1} is given by :

(a) $\begin{bmatrix} 5 & -2 \\ -3 & 1 \end{bmatrix}$ (b) $\begin{bmatrix} -5 & 2 \\ 3 & -1 \end{bmatrix}$

(c) $\begin{bmatrix} -5 & 3 \\ 2 & -1 \end{bmatrix}$ (d) $\begin{bmatrix} 5 & 3 \\ 1 & 2 \end{bmatrix}$

5. In the determinant $\begin{vmatrix} 2 & -3 & 5 \\ 6 & 0 & 4 \\ 1 & 5 & -7 \end{vmatrix}$, M_{23} is :

(where M_{ij} denotes the minor of element a_{ij})

(a) 7 (b) -13

(c) 13 (d) -7

6. If $y = \sec(\tan^{-1} x)$, then $\frac{dy}{dx}$ at $x = 1$ is equal to :

(a) $\sqrt{2}$ (b) $\frac{1}{\sqrt{2}}$

(c) 1 (d) $\frac{1}{2}$

7. The value of k for which the function f given by $f(x) = \begin{cases} \dfrac{k \cos x}{\pi - 2x}, & \text{if } x \neq \dfrac{\pi}{2} \\ 5, & \text{if } x = \dfrac{\pi}{2} \end{cases}$ is continuous at $x = \dfrac{\pi}{2}$, is :

(a) 6 (b) 5

(c) $\dfrac{5}{2}$ (d) 10

8. $\displaystyle\int \frac{dx}{\sin^2 2x \cdot \cos^2 2x}$ equals :

(a) $\dfrac{1}{2}$ [tan 2x + cot 2x] + C (b) tan 2x – cot 2x + C

(c) $\dfrac{1}{2}$ [tan 2x – cot 2x] + C (d) $\dfrac{1}{2}$ [cot 2x – tan 2x] + C

9. $\displaystyle\int_{-\pi/4}^{\pi/4} \sin^3 x \, dx$ equals :

(a) $2 \displaystyle\int_{0}^{\pi/4} \sin^3 x \, dx$ (b) 0

(c) 1 (d) $\displaystyle\int_{0}^{\pi/4} \sin^3 x \, dx$

10. The general solution of the differential equation $\dfrac{dy}{dx} = e^{x-y}$ is :

(a) $e^x + e^y = C$ (b) $e^x - e^{-y} = C$

(c) $-e^x - e^y = C$ (d) $e^x - e^y = C$

11. The sum of the order and the degree of the differential equation $\left(\frac{d^3y}{dx^3}\right)^2 + 3x\left(\frac{d^2y}{dx^2}\right)^4 = \log x$, is :

(a) 5 (b) 6

(c) 7 (d) 4

12. Let $\vec{a}$ and $\vec{b}$ be two unit vectors and θ is the angle between them. $\vec{a} + \vec{b}$ is a unit vector, if :

(a) $\theta = \frac{\pi}{3}$ (b) $\theta = \frac{\pi}{4}$

(c) $\theta = \frac{\pi}{2}$ (d) $\theta = \frac{2\pi}{3}$

13. If $(2\hat{i} + 6\hat{j} - 22\hat{k}) \times (\hat{i} + \lambda\hat{j} + \mu\hat{k}) = \vec{0}$, then $\lambda - \mu$ is equal to :

(a) -8 (b) -14

(c) 14 (d) 8

14. If the position vectors of two points A and B are $\hat{i} + 2\hat{j} - 3\hat{k}$ and $-\hat{i} - 2\hat{j} + \hat{k}$ respectively, then the direction cosines of the vector $\overrightarrow{BA}$ are :

(a) $\frac{2}{6}, -\frac{4}{6}, -\frac{4}{6}$ (b) $\frac{1}{3}, \frac{2}{3}, -\frac{2}{3}$

(c) $\frac{1}{\sqrt{3}}, \frac{2}{\sqrt{3}}, -\frac{2}{\sqrt{3}}$ (d) $-\frac{1}{3}, -\frac{2}{3}, \frac{2}{3}$

15. The value of λ for which the lines $\frac{x-5}{7} = \frac{2-y}{5} = \frac{z}{1}$ and $\frac{x}{1} = \frac{2y-1}{\lambda} = \frac{z}{3}$ are at right angles, is :

(a) 2 (b) 4

(c) -4 (d) -2

16. The solution set of the inequation $2x + y \geq 5$ is :

(a) half plane that contains the origin

(b) open half plane not containing the origin and not containing the points on the line $2x + y = 5$.

(c) whole xy-plane except the points lying on the line $2x + y = 5$.

(d) open half plane not containing the origin, but containing the points on the line $2x + y = 5$.

17. The minimum value of $z = 3x + 8y$ subject to the constraints $x \leq 20$, $y \geq 10$ and $x \geq 0$, $y \geq 0$ is :

(a) 80 (b) 140

(c) 0 (d) 60

18. Two events A and B will be independent, if :

(a) A and B are mutually exclusive

(b) $P(A) = P(B)$

(c) $P(A'B') = [1 - P(A)]\ [1 - P(B)]$

(d) $P(A) + P(B) = 1$

*Questions number **19** and **20** are Assertion and Reason based questions carrying 1 mark each. Two statements are given, one labelled Assertion (A) and the other labelled Reason (R). Select the correct answer from the codes (a), (b), (c) and (d) as given below.*

(a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(b) Both Assertion (A) and Reason (R) are true, but Reason (R) is ***not*** the correct explanation of the Assertion (A).

(c) Assertion (A) is true and Reason (R) is false.

(d) Assertion (A) is false and Reason (R) is true.

19. *Assertion (A) :* Matrix $A = \begin{bmatrix} 0 & -3 & 5 \\ 3 & 0 & -2 \\ -5 & 2 & 0 \end{bmatrix}$ is a skew-symmetric matrix.

Reason (R) : If $A' = -A$, then A is a skew-symmetric matrix.

20. *Assertion (A) :* The vector equation of a line passing through the points $A(-1, 0, 2)$ and $B(3, 4, 6)$ is $\vec{r} = -\hat{i} + 2\hat{k} + \lambda(\hat{i} + \hat{j} + \hat{k})$.

Reason (R) : The equation of a line passing through a point with position vector $\vec{a}$ and parallel to a vector $\vec{b}$, is $\vec{r} = \vec{a} + \lambda\vec{b}$.

SECTION B

This section comprises very short answer (VSA) type questions of 2 marks each.

21. (a) Find the value of $\tan^{-1}\sqrt{3} - \sec^{-1}(-2)$.

OR

(b) Check the injectivity and surjectivity of the function $f : N \rightarrow N$, given by $f(x) = x^3$.

22. If $\cos y = x \cos(a + y)$, and $\cos a \neq \pm 1$, prove that

$$\frac{dy}{dx} = \frac{\cos^2(a+y)}{\sin a}.$$

23. Find the projection of the vector $7\hat{i} - \hat{j} + 8\hat{k}$ on the vector $\hat{i} + 2\hat{j} + 2\hat{k}$.

24. (a) In a parallelogram ABCD, the sides AB and AD are represented by the vectors $2\hat{i} - 4\hat{j} + 5\hat{k}$ and $\hat{i} - 2\hat{j} - 3\hat{k}$ respectively. Find the unit vector parallel to its diagonal $\vec{AC}$.

OR

(b) Find the angle between the pair of lines given by

$\vec{r} = \hat{i} + 2\hat{j} - 2\hat{k} + \lambda(\hat{i} - 2\hat{j} - 2\hat{k})$ and

$\vec{r} = 3\hat{i} - 5\hat{j} + \hat{k} + \mu(3\hat{i} + 2\hat{j} - 6\hat{k})$.

25. The radius of an air bubble is increasing at the rate of 0·5 cm/s. At what rate is the surface area of the bubble increasing when the radius is 1·5 cm ?

SECTION C

This section comprises short answer (SA) type questions of 3 marks each.

26. Find :

$$\int \frac{dx}{1 + \cot x}$$

27. (a) Evaluate :

$$\int_{\pi/2}^{\pi} e^x \left(\frac{1 - \sin x}{1 - \cos x}\right) dx$$

OR

(b) Evaluate :

$$\int_{0}^{\pi/4} \log\,(1 + \tan x)\, dx$$

28. Find :

$$\int \frac{x}{(x^2+1)(x-1)}\, dx$$

29. (a) Find a particular solution of the differential equation $\frac{dy}{dx} + y \cot x = 4x \operatorname{cosec} x$, $(x \neq 0)$, given that $y = 0$ when $x = \frac{\pi}{2}$.

OR

(b) Solve the differential equation $x\, dy - y\, dx = \sqrt{x^2 + y^2}\, dx$.

30. The objective function $z = 4x + 3y$ of a linear programming problem under some constraints is to be maximized and minimized. The corner points of the feasible region are A(0, 700), B(100, 700), C(200, 600) and D(400, 200). Find the point at which z is maximum and the point at which z is minimum. Also, find the corresponding maximum and minimum values of z.

31. (a) A man is known to speak the truth 3 out of 5 times. He throws a pair of different coins and reports that he got a pair of heads. Find the probability that a pair of heads actually occurs.

OR

(b) From a lot of 10 bulbs which includes 2 defectives, a sample of 2 bulbs is drawn at random without replacement. Find the probability distribution of the number of defective bulbs. Hence, find the mean.

SECTION D

This section comprises long answer (LA) type questions of 5 marks each.

32. (a) A relation R in the set A = {5, 6, 7, 8, 9} is given by R = {(x, y) : $|x - y|$ is divisible by 2}. Write R in roster form and prove that R is an equivalence relation. Also, find the elements related to element 7.

OR

(b) Let $A = \mathbb{R} - \{3\}$ and $B = \mathbb{R} - \{1\}$ be two sets. Prove that the function $f : A \rightarrow B$ given by $f(x) = \left(\frac{x-2}{x-3}\right)$ is onto. Is the function f one-one ? Justify your answer.

33. Using matrices, solve the following system of linear equations :

$$x - y + 2z = 7 \; ; \; 3x + 4y - 5z = -5 \; ; \; 2x - y + 3z = 12$$

34. Find the area of the region bounded by the curve $y = \sqrt{x}$, the line $x = 2y + 3$ and the x-axis, using integration.

35. (a) Find the shortest distance between the lines whose vector equations are :

$$\vec{r} = \hat{i} + 2\hat{j} - 4\hat{k} + \lambda(2\hat{i} + 3\hat{j} + 6\hat{k}) \text{ and}$$

$$\vec{r} = 3\hat{i} - 3\hat{j} - 5\hat{k} + \mu(-2\hat{i} + 3\hat{j} + 8\hat{k})$$

OR

(b) Find the foot of the perpendicular drawn from the point (2, 3, – 8) to the line $\frac{x-4}{2} = \frac{y}{-6} = \frac{z-1}{3}$. Also, find the perpendicular distance of the given line from the given point.

SECTION E

This section comprises 3 case study based questions of 4 marks each.

Case Study – 1

36. A balloon is being inflated with the help of an air pump, and it remains spherical. Its radius, the surface area and the volume of air in it are all increasing.

Based on the above, answer the following questions :

(i) Are the quantities : radius, surface area and volume of the spherical balloon changing at the same rate or different rates, when air is filled in it ? *1*

(ii) Write the expressions for the surface area (S) and the volume (V) of the balloon at any time 't' in terms of radius 'r' at that instant. *1*

(iii) (a) At the instant when the radius of the balloon is 6 cm and the radius (r) is increasing at the rate of 2 cm/s, find at what rate the surface area (S) of the balloon is increasing. *2*

OR

(iii) (b) At the instant when the radius of the balloon is 6 cm and the radius (r) is increasing at the rate of 2 cm/s, find at what rate the volume (V) of the spherical balloon is increasing. *2*

Case Study – 2

37. A fighter-jet of the enemy is flying along the parabolic path $4y = x^2$. A soldier is located at the point (0, 5) and is aiming to shoot down the jet when it is nearest to him.

Based on the above, answer the following questions :

(i) Let (x, y) be the position of the jet at any instant. Express the distance between the soldier and the jet as the function f(x). *1*

(ii) Taking $S = [f(x)]^2$, find $\frac{dS}{dx}$. *1*

(iii) (a) What will be the position of the jet when the soldier shoots it down ? *2*

OR

(iii) (b) What will be the distance between the soldier and the jet at the instant when he shoots it down ? *2*

Case Study – 3

38. Read the following passage and answer the questions given below :

There are ten cards numbered 1 to 10 and they are placed in a box and then mixed up thoroughly. Then one card is drawn at random from the box.

Based on the above, answer the following questions :

(i) What is the probability that the number on the drawn card is greater than 4 ? *2*

(ii) If it is known that the number on the drawn card is greater than 4, then what is the probability that it is an even number ? *2*